Construction of Real Numbers

Purpose of these notes is to explain the Dedekind construction of real numbers from the field of rational numbers.

Theorem : There exists a complete ordered field containing the field $\mathbb{Q}$ of rational numbers as a subfield. This complete ordered field is called the field $\mathbb{R}$ of real numbers.

We assume the existence of the field $\mathbb{Q}$ of rational numbers, which is the smallest ordered field. Let $\mathcal{P}(\mathbb{Q})$ be the power set of $\mathbb{Q}$. A subset $\alpha \subset \mathbb{Q}$ is called a *Dedekind cut* (or simply a cut) if the following conditions hold.

1. α is a nonempty proper subset of $\mathbb{Q}$.
2. If $p, q \in \mathbb{Q}, p \in \alpha$ and $q < p$, then $q \in \alpha$.
3. If $p \in \alpha$, then there exists $r \in \alpha$ with $r > p$.

The above properties show that a Dedekind cut α is a nonempty proper subset of $\mathbb{Q}$, which is downward closed and has no largest element. Note that if p lies in a cut α and $s \in \mathbb{Q}$ does not lie in α, then $p < s$.

In what follows, we use $\alpha, \beta, \gamma, \delta, \ldots$ to denote cuts, while $p, q, r, s, \ldots$ denote rational numbers.

Proof. Let $\mathbb{R}$ be the set of all Dedekind cuts. For each rational number $p \in \mathbb{Q}$, there is a cut p^* given by

$$p^* = \{r \in \mathbb{Q} : r < p\}.$$

In other words, p^* is the set of all rational numbers strictly less than p. Clearly, $p^* \in \mathbb{R}$ for each $p \in \mathbb{Q}$. Also, $p < q$ in $\mathbb{Q}$ if and only if $p^* \subsetneq q^*$.

We define a strict order $<$ on $\mathbb{R}$ as follows. For $\alpha, \beta \in \mathbb{R}$,

$$\alpha < \beta \quad \text{if} \quad \alpha \subsetneq \beta.$$

Clearly, $\alpha \not\subset \alpha$. Also, $\alpha \subsetneq \beta$ and $\beta \subsetneq \gamma$, implies that $\alpha \subsetneq \gamma$. This shows that $\alpha \not< \alpha$ for every $\alpha \in \mathbb{R}$ and $<$ is transitive.

(**Trichotomy**) The strict order $<$ on $\mathbb{R}$ satisfies trichotomy property. For $\alpha, \beta \in \mathbb{R}$, one and only one of the three options hold :

$$\alpha < \beta, \quad \text{or} \quad \alpha = \beta, \quad \text{or} \quad \beta < \alpha.$$

If the first two does not hold, then we shall show that the last option is valid. In other words, if $\alpha \nsubseteq \beta$, then we shall show that $\beta \subsetneq \alpha$. As $\alpha \nsubseteq \beta$, there exists $p \in \alpha$ such that $p \notin \beta$. Now for every $q \in \beta$, we have $q < p$. Further, as α is a cut and $q < p$, we have

$q \in \alpha$. Hence $\beta \subsetneq \alpha$.

(**Dedekind Completeness**) The ordered set $(\mathbb{R}, <)$ is Dedekind complete. In other words, every nonempty bounded above subset $\mathcal{A}$ of $\mathbb{R}$ has a supremum in $\mathbb{R}$. Let β be an upper bound of $\mathcal{A}$. Then every $\alpha \in \mathcal{A}$ satisfies $\alpha \subseteq \beta$. Also $\mathcal{A}$ is nonempty. Let $\beta^* = \bigcup \mathcal{A} = \bigcup\{\alpha : \alpha \in \mathcal{A}\}$. Cleary, every $\alpha \in \mathcal{A}$ is contained in β^* and β^* is a nonempty subset of $\mathbb{Q}$. Now $\beta^* = \bigcup \mathcal{A} \subseteq \beta$ implies that β^* is an upper bound of the set $\mathcal{A}$ and being the union of all the members of $\mathcal{A}$, it is the smallest subset of $\mathbb{Q}$ containing all the members of $\mathcal{A}$. If we show that β^* is a cut, then indeed, it will be the supremum of $\mathcal{A}$.

As $\mathcal{A}$ is nonempty subset of $\mathbb{R}$ with an upper bound β, we see that β^* is nonempty subset contained in β. Thus β^* is a nonempty proper subset of $\mathbb{Q}$. Let $p, q, r \in \mathbb{Q}$ with $q < p < r$. If $p \in \beta^*$, then there exists $\alpha \in \mathcal{A}$ such that $p \in \alpha$. As α is a cut, $q \in \alpha$ for every $q < p$. Hence, $q \in \bigcup \mathcal{A} = \beta^*$. Also, there exists some $r > p$ such that $r \in \alpha$. Again, $r \in \beta^*$. Hence, β^* is a cut, as desired.

(**Additive Structure on** $\mathbb{R}$) For $\alpha, \beta \in \mathbb{R}$, define the sum $\alpha + \beta$ as

$$\alpha + \beta = \{p + q : p \in \alpha, q \in \beta\}.$$

We need to verify that $\alpha + \beta$ is a cut. Clearly, there are $p_0 \in \alpha$ and $q_0 \in \beta$. Also, there are $r_0, s_0 \in \mathbb{Q}$ such that $r_0 \notin \alpha$ and $s_0 \notin \beta$. Thus $p_0 + q_0 \in \alpha + \beta$ and $r_0 + s_0 \notin \alpha + \beta$, hence $\alpha + \beta$ is a nonempty proper subset of $\mathbb{Q}$. Let $p \in \alpha$, $q \in \beta$ and $s < p + q$ be a rational. Then $s - p < q$ implies that $s - p \in \beta$. Hence $s = p + (s - p) \in \alpha + \beta$. This shows that $\alpha + \beta$ is downward closed. Finally, if $p + q \in \alpha + \beta$ with $p \in \alpha$ and $q \in \beta$. Then there is an $r > p$ such that $r \in \alpha$. Clearly, $r + q > p + q$ and $r + q \in \alpha + \beta$. This shows that $\alpha + \beta$ is a cut.

(**Addition is Commutative and Associative**) Consider $\alpha, \beta, \gamma \in \mathbb{R}$. As $p + q = q + p$ for every $p, q \in \mathbb{Q}$, we see that

$$\alpha + \beta = \{p + q : p \in \alpha, q \in \beta\} = \beta + \alpha.$$

Also, $p + (q + r) = (p + q) + r$ for every $p, q, r \in \mathbb{Q}$. Hence,

$$\alpha + (\beta + \gamma) = \{p + (q + r) : p \in \alpha, q \in \beta, r \in \gamma\} = (\alpha + \beta) + \gamma.$$

(0^* **is the Additive Identity**) Consider the cut $0^* = \{q \in \mathbb{Q} : q < 0\}$ determined by the rational number 0. Let $\alpha \in \mathbb{R}$. We need to show that

$$\alpha + 0^* = \alpha.$$

If $p \in \alpha$ and $q \in 0^*$, then $p + q < p$ as $q < 0$. Since α is downward closed, $p + q \in \alpha$. Thus, $\alpha + 0^* \subseteq \alpha$. Conversely, let $p \in \alpha$, then there is an $r > p$ with $r \in \alpha$. Now $p - r < 0$ implies that $p = r + (p - r) \in \alpha + 0^*$. This shows that $\alpha \subseteq \alpha + 0^*$. Hence, $\alpha + 0^* = \alpha$.

(**Additive Inverse**) Let $\alpha \in \mathbb{R}$. Define a subset $\gamma \subseteq \mathbb{Q}$ by

$$\gamma = \{p \in \mathbb{Q} : \exists\, r > 0 \text{ with} - p - r \notin \alpha\}.$$

Geometrically, γ is a subset of $\mathbb{Q}$ obtained by the reflection of $\mathbb{Q} \setminus \alpha$ through origin and if $-(\mathbb{Q} \setminus \alpha)$ has the largest number $\ell \in \mathbb{Q}$, then $\gamma = -(\mathbb{Q} \setminus \alpha) \setminus \{\ell\}$, otherwise $\gamma = -(\mathbb{Q} \setminus \alpha)$.

First, we show that γ is a cut. Let $p \notin \alpha$. Then $p + 1 \notin \alpha$. Thus, $r = -(p+1) \in \gamma$. Hence, γ is nonempty. Further, $p \in \alpha$ implies that $-p \notin \gamma$. For if, $-p \in \gamma$, then there exists $r > 0$ such that $p - r \notin \alpha$, a contadiction as $p \in \alpha$ and $p - r < p$. This shows that γ is a proper subset of $\mathbb{Q}$. Let $p \in \gamma$ and $q < p$. Then $\exists r > 0$ with $-p - r \notin \alpha$. As $-p - r < -q - r$, we have $-q - r \notin \alpha$. Therefore, $q \in \gamma$. Also, for $p \in \gamma$ as in the last line, we see that $p + \frac{r}{2} \in \gamma$, becasue $-(p + \frac{r}{2}) - \frac{r}{2} \notin \alpha$. This proves that γ is a cut.

We now show that

$$\gamma + \alpha = 0^*.$$

Let $p \in \gamma$ and $q \in \alpha$. Then there exists $r > 0$ such that $-p - r \notin \alpha$. As α is a cut, $q < -p - r$. Equivalently, $p + q < -r < 0$ and $\gamma + \alpha \subseteq 0^*$. Let $-s \in 0^*$. Then s is a positive rational number. Since α is a cut, there exists $p \in \alpha$ and $q \notin \alpha$. Since $\mathbb{Q}$ is an Archimedean field, there is an integers n such that $q < p + ns$. Therefore, $p + ns \notin \alpha$. Choose smallest positive integer n_0 such that $p + n_0 s \notin \alpha$ but $p + (n_0 - 1)s \in \alpha$. Choose $p' \in \alpha$ with $p' > p + (n_0 - 1)s$, then $p' + s \notin \alpha$. Taking $r = \frac{1}{2}(p' - p - (n_0 - 1)s) > 0$, we see that $p' + s - 2r = p + n_0 s$ or $p' + s - r > p + n_0 s$. This shows that $-p' - s \in \gamma$. Now $(-p' - s) + p' = -s \in \gamma + \alpha$, showing that $0^* \subseteq \gamma + \alpha$. Hence, $\gamma + \alpha = 0^*$.

The cut γ is the additive inverse of α, we denote this cut by $-\alpha$.

Thus $\mathbb{R}$ has a structure of an Abelian group with respect to addition. As every additive group has a unique additive identity and every element has a unique additive inverse, we see that if $\alpha, \beta \in \mathbb{R}$, then the additive inverse of $\alpha + \beta$ is given by

$$-(\alpha + \beta) = (-\alpha) + (-\beta) = -\alpha - \beta.$$

(**Multiplicative Structure on** $\mathbb{R}$) Multiplication of cuts is first defined for positive cuts. A cut α is positive i.e., $\alpha > 0^*$ if $0 \in \alpha$. Consider two positive cuts α and β. Then we define the product $\alpha\beta$ by

$$\alpha\beta = \{q' \in \mathbb{Q} : q' \leq 0\} \cup \{pq : p > 0, q > 0 \text{ and } p \in \alpha, q \in \beta\}.$$

We shall show that $\alpha\beta$ is a cut. As $0^* \subset \alpha\beta$, it is nonempty. Also, if $p_0 \notin \alpha$ and $q_0 \notin \beta$, then $p_0 q_0 \notin \alpha\beta$. This shows that $\alpha\beta$ is a proper subset of $\mathbb{Q}$. Let $pq \in \alpha\beta$ with $0 < p \in \alpha$ and $0 < q \in \beta$. If $0 < p' < pq$, then $p'q^{-1} < p$ implies that $p'q^{-1} \in \alpha$. Hence, $p' = (p'q^{-1})q \in \alpha\beta$. Also, for $pq \in \alpha\beta$ as in the last lines, there exists $r > p$ with $r \in \alpha$. Clearly, $pq < rq \in \alpha\beta$. This proves that $\alpha\beta$ is a cut.

Let $\mathbb{R}^+ = \{\alpha \in \mathbb{R} : 0 \in \alpha\}$ be the set of all positive Dedekind cuts.

(**Mutliplication is Commutative and Associative**) Let $\alpha, \beta, \gamma \in \mathbb{R}^+$. As product of rational numbers is both commutative and associative, we see that

$$\alpha\beta = \{q' \leq 0\} \cup \{pq : 0 < p \in \alpha, 0 < q \in \beta\} = \beta\alpha.$$

Also,

$$\alpha(\beta\gamma) = \{q' \leq 0\} \cup \{p(qr) : 0 < p \in \alpha, 0 < q \in \beta, 0 < r \in \gamma\} = (\alpha\beta)\gamma.$$

(1^* **is the Multiplicative Identity**) Consider the cut $1^* = \{p \in \mathbb{Q} : p < 1\}$ determined by the rational number 1. We shall show that for $\alpha \in \mathbb{R}^+$,

$$\alpha 1^* = \alpha.$$

Let $0 < p \in \alpha$ and $0 < q < 1$, then $pq < p$ implies that $pq \in \alpha$. Thus $\alpha 1^* \subseteq \alpha$. Let $0 < p \in \alpha$. Then there is an $r \in \alpha$ with $r > p$. Clearly, $0 < \frac{p}{r} < 1$ and $p = r(\frac{p}{r}) \in \alpha 1^*$. Thus $\alpha \subseteq \alpha 1^*$ and hence $\alpha 1^* = \alpha$ This shows that 1^* is the multiplicative identity for $\mathbb{R}^+$.

(**Multiplicative Inverse**) Let $\alpha \in \mathbb{R}^+$. Define a subset $\delta \subsetneq \mathbb{Q}$ by

$$\delta = \{q' \in \mathbb{Q} : q \leq 0\} \cup \{p \in \mathbb{Q} : p > 0 \text{ and } \exists\ r > 0 \text{ with } (p^{-1}) - r \notin \alpha\}.$$

Geometrically, δ is described as the union of $\{q' \leq 0\}$ with the inversion of the subset $\mathbb{Q} \setminus \alpha$ and if t is the largest element of $(\mathbb{Q} \setminus \alpha)^{-1} = \{0 < p^{-1} \in \mathbb{Q} : p \notin \alpha\}$, then $\delta = \{q' \leq 0\} \cup ((\mathbb{Q} \setminus \alpha)^{-1} \setminus \{t\})$, otherwise $\delta = \{q' \leq 0\} \cup (\mathbb{Q} \setminus \alpha)^{-1}$.

We shall show that δ is a cut. Clearly, $0 \in \delta$ and if $0 < p \in \alpha$, then $p^{-1} \notin \delta$. For if, $0 < p^{-1} \in \delta$, then for every $r > 0$, we have $p - r \in \alpha$ as $p - r < p \in \alpha$. This shows that δ is a nonempty proper subset of $\mathbb{Q}$. If $0 < p \in \delta$ and $0 < q < p$, then $\exists\ r > 0$ such that $p^{-1} - r \notin \alpha$. Now $p^{-1} - r < q^{-1} - r$ implies that $q^{-1} - r \notin \alpha$ and hence $q \in \delta$. This shows that δ is downward closed. For $0 < p \in \delta$ as in the last line, then consider $\tilde{p} > 0$ such that $\frac{1}{\tilde{p}} = \frac{1}{p} - \frac{r}{2}$. Clearly, $p < \tilde{p}$ and $\tilde{p} \in \delta$ as $\frac{1}{\tilde{p}} - \frac{r}{2} = \frac{1}{p} - r \notin \alpha$. Thus δ is a cut.

Now we shall show that

$$\delta\alpha = 1^*.$$

Let $0 < p \in \delta$ and $0 < q \in \alpha$. Then $\exists\ r > 0$ with $p^{-1} - r \notin \alpha$. Thus $q < p^{-1} - r < p^{-1}$. Hence, $pq < 1$. This shows that $\delta\alpha \subseteq 1^*$. Conversely, let $0 < s < 1$. Now consider $ps \notin \alpha$. Since s^n tends to 0 as n tends to infinity, there exists a positive integer m such that $ps^m \notin \alpha$ and $ps^{m+1} \in \alpha$. Choose $p' \in \alpha$ with $p' > ps^{m+1}$. Then $\frac{p'}{s} > ps^m$. This shows that $\frac{s}{p'} \in \delta$. Hence $s = \left(\frac{s}{p'}\right) p' \in \delta\alpha$. Therefore, $1^* \subseteq \delta\alpha$ and $\delta\alpha = 1^*$.

The cut δ is the multiplicative inverse of $\alpha \in \mathbb{R}^+$ and we denote it with α^{-1}.

This shows that $\mathbb{R}^+$ is a multiplicative Abelian group with multiplicative identity 1^*.

We now extend multiplication to all cuts in $\mathbb{R}$. If $\alpha \in \mathbb{R}$, then we define

$$\alpha 0^* = 0^* \alpha = 0^*.$$

Let $\alpha, \beta \in \mathbb{R}$ with $0^* \neq \alpha$ and $0^* \neq \beta$. If $\alpha > 0^*$ and $\beta > 0^*$, then the multiplication $\alpha\beta$ has already been defined. If $\alpha > 0^*$ and $\beta < 0^*$, then we define $\alpha\beta = -(\alpha(-\beta))$. If $\alpha < 0^*$ and $\beta > 0^*$, then we define $\alpha\beta = -((-\alpha)\beta)$. Finally, if $\alpha < 0^*$ and $\beta < 0^*$, then we define $\alpha\beta = (-\alpha)(-\beta)$.

Now for $0^* \neq \alpha \in \mathbb{R}$, we show that it has a multiplicative inverse. If $\alpha > 0^*$, we have already obtained its multiplicative inverse α^{-1}. If $\alpha < 0^*$, then the multiplicative inverse $\alpha^{-1} = -(-\alpha)^{-1}$.

(**Distributive Law**) For $\alpha, \beta, \gamma \in \mathbb{R}$, we shall prove that

$$\alpha(\beta + \gamma) = \alpha\beta + \alpha\gamma.$$

If any one of α, β or γ is 0^*, then $\alpha(\beta+\gamma) = \alpha\beta + \alpha\gamma$ holds. If $\alpha = 0^*$, then both sides are zero. If either $\beta = 0^*$ or $\gamma = 0^*$, then accordingly either both sides are $\alpha\gamma$ or both sides are $\alpha\beta$. Thus we may assume that all α, β and γ are different from 0^*. Further, if $\beta + \gamma = 0^*$, then $\beta = -\gamma$ and we see that both sides are equal to 0^*.

Case 1. Suppose $\alpha \neq 0^*, \beta > 0^*, \gamma > 0^*$. If $\alpha > 0^*$, then

$$\alpha(\beta + \gamma) = \{q' \leq 0\} \cup \{p(q + r) : p, q, r \in \mathbb{Q}^+ \text{ with } p \in \alpha, q \in \beta, r \in \gamma\}.$$

For $p, q, r \in \mathbb{Q}^+$, we have $p(q + r) = pq + pr$. Thus

$$\alpha\beta + \alpha\gamma = \{q' \leq 0\} \cup \{pq + pr) : p, q, r \in \mathbb{Q}^+ \text{ with } p \in \alpha, q \in \beta, r \in \gamma\} = \alpha(\beta + \gamma).$$

Now, suppose $\alpha < 0^*$. Then $\alpha_1 = -\alpha > 0^*$. Also,

$$\alpha(\beta + \gamma) = -(\alpha_1(\beta + \gamma)) \quad \text{and} \quad \alpha\beta + \alpha\gamma = -(\alpha_1\beta) - (\alpha_1\gamma).$$

From the first part, $\alpha_1(\beta + \gamma) = \alpha_1\beta + \alpha_1\gamma$. Also, $-(\alpha_1\beta + \alpha_1\gamma) = -(\alpha_1\beta) - (\alpha_1\gamma)$. Therefore, for $\alpha < 0^*$, we also have $\alpha(\beta + \gamma) = \alpha\beta + \alpha\gamma$.

Case 2. Suppose $\alpha \neq 0^*, \beta < 0^*, \gamma < 0^*$. Assume $\alpha > 0^*$. Let $\beta_1 = -\beta > 0^*$ and $\gamma_1 = -\gamma > 0^*$. Then $\beta_1 + \gamma_1 = -(\beta + \gamma) > 0^*$. Thus we have

$$\alpha(\beta + \gamma) = -(\alpha(\beta_1 + \gamma_1)) \quad \text{and} \quad \alpha\beta + \alpha\gamma = -(\alpha\beta_1) - (\alpha\gamma_1).$$

From Case 1, $\alpha(\beta_1 + \gamma_1) = \alpha\beta_1 + \alpha\gamma_1$. Also, $-(\alpha\beta_1 + \alpha\gamma_1) = -(\alpha\beta_1) - (\alpha\gamma_1)$. This shows that for $\alpha > 0^*$, we have $\alpha(\beta + \gamma) = \alpha\beta + \alpha\gamma$.

Now assume that $\alpha < 0^*$ and $\alpha_1 = -\alpha > 0^*$. Then

$$\alpha(\beta + \gamma) = \alpha_1(\beta_1 + \gamma_1) \quad \text{and} \quad \alpha\beta + \alpha\gamma = \alpha_1\beta_1 + \alpha_1\gamma_1.$$

Again by Case 1, both sides are the same.

Case 3. Suppose $\alpha \neq 0^*$ and $\beta\gamma < 0^*$, i.e. β and γ have different signs. Assume that $\beta > 0^*$ and $\gamma < 0^*$, otherwise interchange their roles. We have already seen that distributive law holds if $\beta + \gamma = 0^*$. Thus assume that $\beta + \gamma \neq 0^*$.

If $\beta+\gamma > 0^*$, then $\gamma_1 = -\gamma > 0^*$ and $\beta = (\beta+\gamma)+\gamma_1$. Now by Case 1, we have

$$\alpha\beta = \alpha((\beta+\gamma)+\gamma_1) = (\alpha(\beta+\gamma)) + \alpha\gamma_1.$$

Since $\alpha\gamma + \alpha\gamma_1 = 0^*$, we get $\alpha\beta + \alpha\gamma = \alpha(\beta+\gamma)$, as desired.

If $\beta+\gamma < 0^*$, then $\beta_1 = -\beta < 0^*$ and $\gamma = (\beta+\gamma)+\beta_1$. Now by Case 2, we have

$$\alpha\gamma = \alpha((\beta+\gamma)+\beta_1) = (\alpha(\beta+\gamma)) + \alpha\beta_1.$$

Since $\alpha\beta + \alpha\beta_1 = 0^*$, we again get $\alpha\beta + \alpha\gamma = \alpha(\beta+\gamma)$. This completes the prove of distributive law. Hence we have shown that $(\mathbb{R},\ +,\ \cdot,\ 0^*,\ 1^*)$ is a field.

($\mathbb{R}$ **is an Ordered Field**) We have already constructed a strict order $<$ on $\mathbb{R}$ and proved that trichotomy holds. We need only to show that if $\alpha > 0^*$ and $\beta > 0^*$, then $\alpha+\beta > 0^*$ and $\alpha\beta > 0^*$. Noe that $\alpha > 0^*$ if and only if $0 \in \alpha$. Clearly, $0 \in \alpha+\beta$ and by definition of product of positive cuts, we have $0 \in \alpha\beta$. This shows that $\mathbb{R}$ is an ordered field. We have already proved Dedekind completeness of $\mathbb{R}$ and hence, $\mathbb{R}$ is a complete ordered field.

($\mathbb{R}$ **Contains** $\mathbb{Q}$ **as an Ordered Field**) For $p \in \mathbb{Q}$, the rational cut $p^* \in \mathbb{R}$ is given by $p^* = \{p' \in \mathbb{Q} : p' < p\}$. For $p, q \in \mathbb{Q}$, we have already observed that $p < q$ if and only if $p^* < q^*$. Thus we need to verify that

$$(p+q)^* = p^* + q^* \quad \text{and} \quad (pq)^* = p^*q^*.$$

Let $p'+q' \in p^*+q^*$ with $p' < p$ and $q' < p$. Then $p'+q' < p+q$ implies that $p'+q' \in (p+q)^*$. Thus $p^*+q^* \subseteq (p+q)^*$. Conversely, let $\ell \in (p+q)^*$. Then $\ell < p+q$. Clearly, $\ell - q < p$. Choose $p' \in \mathbb{Q}$ with $\ell - q < p' < p$. Then $\ell - p' < q$ and we see that $\ell = p' + (\ell - p') \in p^* + q^*$. Thus $(p+q)^* \subseteq p^*+q^*$ and hence $(p+q)^* = p^*+q^*$.

From $(p+q)^* = p^*+q^*$, we see that $(-p)^* = -(p^*)$ for every $p > 0$. Thus we only need to verify $(pq)^* = p^*q^*$ for $p > 0, q > 0$. Since both sides contains the set $\{r \in \mathbb{Q} : r \leq 0\}$, we take $p'q' \in p^*q^*$ with $0 < p' < p$ and $0 < q' < q$. Clearly, $0 < p'q' < pq$ implies that $p'q' \in (pq)^*$. Thus $p^*q^* \subseteq (pq)^*$. Conversely, let $\ell \in (pq)^*$ with $0 < \ell < pq$. We see that $0 < \ell q^{-1} < p$ and we choose p' with $0 < \ell q^{-1} < p' < p$. Thus $0 < \ell(p')^{-1} < q$ and $\ell = p'(\ell(p')^{-1}) \in p^*q^*$. This shows that $(pq)^* \subseteq p^*q^*$ and $(pq)^* = p^*q^*$.

We consider the subset $\mathbb{Q}^* = \{p^* \in \mathbb{R} : p \in \mathbb{Q}\}$ of $\mathbb{R}$ consisting of rational cuts. Then $\mathbb{Q}^*$ is an ordered subfield of $\mathbb{R}$ isomorphic with the ordered field $\mathbb{Q}$ of rational numbers.

Exercise : Let $(\mathbb{K}, <')$ be a complete ordered field and $j : \mathbb{Q} \to \mathbb{K}$ be a copy of the ordered field $\mathbb{Q}$ of rational numbers in $\mathbb{K}$. Then show that there is a unique field isomorphism $f : \mathbb{R} \to \mathbb{K}$, which is order-preserving.

Hints: For $x \in \mathbb{R}$, set $a_n = \frac{\lfloor 2^n x \rfloor}{2^n}$ for $n \in \mathbb{N}$. Then (a_n) is a convergent sequence of rational numbers such that $x = \lim_{n\to\infty} a_n$. Define $f(x) = \lim_{n\to\infty} j(a_n) \in \mathbb{K}$. Verify that f has the required properties.